223 results for “survey review wildfire ”forest fire” “machine learning” ”artificial intelligence” ”deep learning” techniques management”

Fields of Study | Date Range | Has PDF | Author | Journals & Conferences | More Filters | Filters | Sort by Relevance

Mapping Homogeneous Response Areas for Forest Fuel Management Using Geospatial Data, K-Means, and Random Forest Classification

Á. Chávez-Durán | M. Olvera-Vargas | B. Figueroa-Rangel | Mariano García | I. Aguado | J. A. Ruiz-Corral

Environmental Science, Engineering · Forests · 22 November 2022

Accurate description of **forest** fuels is necessary for developing appropriate **fire management** strategies aimed at reducing **fire** risk. Although field **surveys** provide accurate measurements of **forest**… Expand

8 | PDF | Publisher | Save | Cite

A Hybrid Framework for Forest Fire Detection and Severity Prediction using Sequential Deep Learning on Multitemporal Satellite Imagery

Rohini Bhosale | Poonam Railkar | Environmental Science, Computer Science ·

EPJ Web of Conferences · 2025

TLDR A **deep learning**-based hybrid approach using change detection, Long Short-Term Memory (LSTM) and attention mechanism on pre-processed satellite images is proposed for the early detection of **forest fires**, demonstrating high accuracy, surpassing traditional methods, and aids in early warning and decision-making for **fire management** authorities. Expand

Publisher | Save | Cite

Explainable AI Based Machine Learning Approaches for Power Transformer Fault Diagnosis Methods

V. Chavan | Dr. Mithilesh Singh | Professor Head | Dr. P. William | Engineering, Computer Science ·

2025 International Conference on **Artificial**… · 19 December 2025

TLDR This paper **surveys** XAI-based ML approaches applied to transformer diagnostics, covering feature engineering from multimodal sensor inputs, supervised and semi-supervised models, and post-hoc and intrinsic explainability **techniques**. Expand

IEEE | Save | Cite

The NeurIPS '18 Competition: From Machine Learning to Intelligent Conversations

Computer Science · The Springer Series on Challenges in **Machine**… · 2020

TLDR This volume presents the details of eight competitions in the area of AI which took place between February to December 2018 and were presented at the Neural Information Processing Systems conference in Montreal, Canada on December 8, 2018. Expand

15 | PDF | Publisher | Save | Cite

FathomNet: A global underwater image training set for enabling artificial intelligence in the ocean

K. Katija | Eric C. Orenstein | +6 authors | K. L. Bell | Environmental Science, Computer Science ·

arXiv.org · 2021

TLDR FathomNet is built, a public platform that makes use of existing, expertly curated data and demonstrates how **machine learning** models trained on FathomNet data can be applied across different institutional video data, and enable automated acquisition and tracking of midwater animals using a remotely operated vehicle. Expand

7 | Save | Cite

Ai Technology Achieving General Purpose Ai That Can

Computer Science, Education · 2022

TLDR This book constitutes the refereed proceedings of the Second International Conference, SLAAI-ICAI 2018, held in Moratuwa, Sri Lanka, in December 2018 and contains 32 revised full papers presented. Expand

Save Cite

Multi Uav Cooperative Surveillance With Spatio Temporal

Computer Science, Engineering · 2022

TLDR A wide-ranging **review** of the latest developments in the cooperative control of multi-agent systems theory and applications in the areas of unmanned aerial vehicles (UAVs) and unmanned ground vehicles (UGVs). Expand

Save Cite

A review on the applications of machine learning and deep learning to groundwater salinity modeling: present status, challenges, and future directions

Dilip Kumar Roy Tapash Kumar Sarkar Tasnia Hossain Munmun Chitra Paul B. Datta

Environmental Science, Computer Science · Discover Water · 27 February 2025

TLDR The **review** highlights the prospects of ML, DL, and ensemble models to redefine groundwater **management** in coastal aquifers, offering more effective and sustainable solutions to the challenges posed by groundwater salinity and saltwater intrusion. Expand

9 Springer Nature Save Cite

A Comprehensive Review of Machine Learning and Deep Learning Methods for Flood Inundation Mapping

Abinash Silwal Anil Subedi +4 authors Mohamed Zhran

Environmental Science, Computer Science · Earth · 9 March 2026

TLDR This **review** synthesizes recent advances in ML-based flood inundation mapping, categorizing methods into traditional **machine learning techniques**, **deep learning** architectures (e.g., Random **Forest**, Support Vector **Machines**, Gradient Boosting), and emerging hybrid and physics-informed frameworks. Expand

Publisher Save Cite

Semantic speech analysis using machine learning and deep learning techniques: a comprehensive review

Suryakant Tyagi S. Szénási Computer Science · Multimedia tools and applications · 19 December 2023

TLDR This study analyzes how emotional chatbots, facial expressions, images, and social media texts can be effective in detecting emotions and shows that Long-Short Term Memory (LSTM) and Convolutional Neural Networks (CNN) are found to be the most useful methodology for this purpose. Expand

8 PDF Springer Nature Save Cite

← ... 16 17 18 19 ... →

Stay Connected With Semantic Scholar

Your E-mail Address Sign Up

What Is Semantic Scholar?

Semantic Scholar is a free, AI-powered research tool for scientific literature, based at Ai2.

Learn More

About

About Us
Publishers
Blog
Ai2 Careers

Product

Product Overview
Semantic Reader
Scholar's Hub
Beta Program
Release Notes

API

API Overview
API Tutorials
API Documentation
API Gallery

Research

Publications
Research Careers
Resources

Help

FAQ
Librarians
Tutorials
Contact

Proudly built by Ai2
Collaborators & Attributions • Terms of Service • Privacy Policy • API License Agreement

Ai2